Assignment 3

STAT 8670

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Contents

1 Question 1

Let $\theta = \int\limits_2^4 e^{-x} dx$

1.1 1a

Derive θ in a closed form.

□

$$\theta = \int_2^4 e^{-x} dx = -e^{-x}\Big|_2^4 = (-e^{-4}) - (-e^{-2}) = e^{-2} - e^{-4}$$

■

1.2 1b

Consider the simple MC estimator with $X_i \sim \text{Unif}(2, 4)$

$\hat{\theta}_m = (4-2)\frac{1}{m}\sum_{i=1}^{m} \mathrm{e}^{-X_i}$

Derive $\mathbb{E}[\hat{\theta}_m], \text{Var}(\hat{\theta}_m)$ and a large-sample $100(1-\alpha)\%$ CI for θ using the CLT

□

$$\mathbb{E}[\hat{\theta}_m] = 2\mathbb{E}[e^{-X}] = 2\int_2^4 e^{-x} f_{\text{U}(2,4)} dx = \int_2^4 e^{-x} dx = \theta$$

$$\text{Var}(\hat{\theta}_m) = 4\frac{\text{Var}(e^{-X})}{m} = \frac{4}{m}\left(\mathbb{E}\left[e^{-2X}\right] - \mathbb{E}\left[e^{-X}\right]^2\right) = \frac{4}{m}\left(\frac{e^{-4} - e^{-8} - \left(e^{-2} - e^{-4}\right)^2}{4}\right)$$

$$\text{Var}(\hat{\theta}_m) = \frac{e^{-4} - e^{-8} - \left(e^{-2} - e^{-4}\right)^2}{m} = \frac{2(e^{-6} - e^{-8})}{m}$$

By the large sample CLT, we have $\hat{\theta}_m \approx N(\theta, \text{Var}(\hat{\theta}_m))$. So we can approximate this Ci by the following.

$$\hat{\theta}_m \pm z_{1-\alpha/2}\sqrt{\frac{2\left(e^{-6} - e^{-8}\right)}{m}}$$

■

1.3 1c

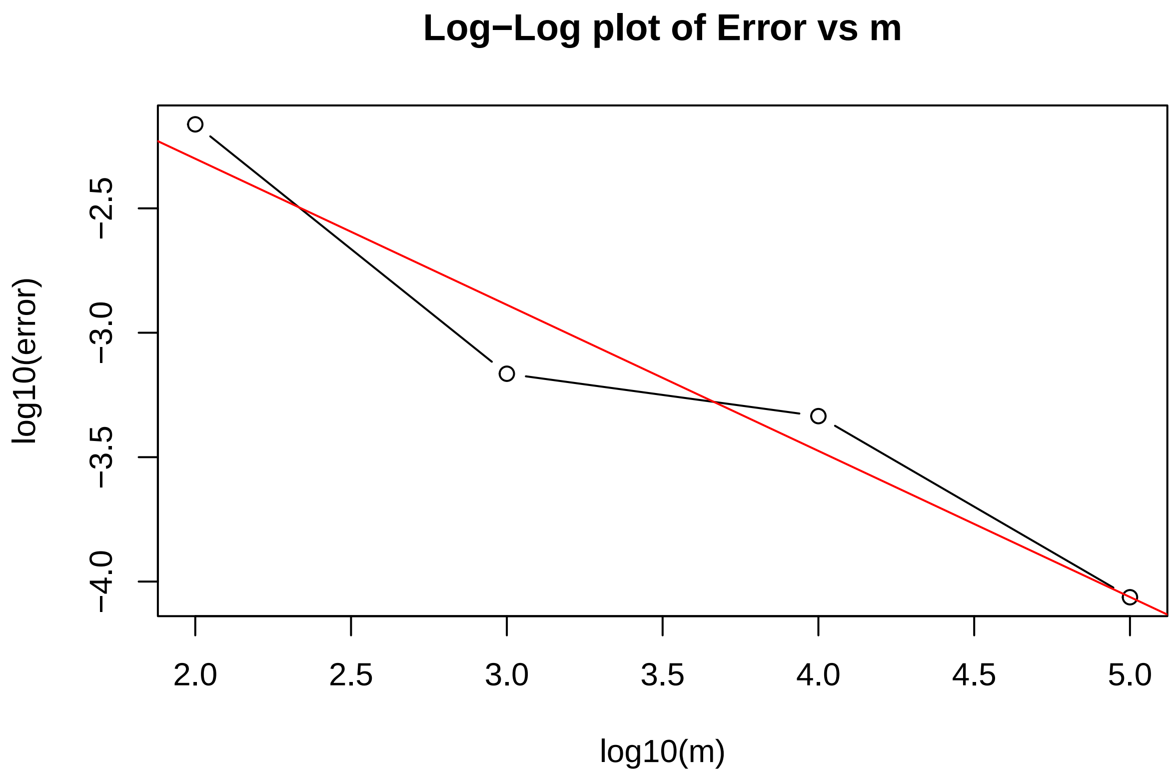
Simulate $m \in \{10^2, 10^3, 10^4, 10^5\}$ and produce a log-log plot of absolute error vs m with setting seeds. Fit a least-squares line to $\log_{10}$(abs. error) vs $\log_{10}(m)$ and report the slope.

□

```r
theta_true <- exp(-2) - exp(-4)

m_samples <- c(1e2, 1e3, 1e4, 1e5)

theta_sample <- function(m){

    running_sum <- 0

    for (i in 1:m){
        set.seed(i)
        x <- runif(1, 2, 4)
        running_sum <- running_sum + exp(-x)
    }

    return((4-2)*running_sum / m)
}

m_sampled_values <- numeric(length(m_samples))

for (i in seq_along(m_samples)){
    m <- m_samples[i]
    m_sampled_values[i] <- theta_sample(m)
}

m_sampled_errors <- abs(m_sampled_values - theta_true)
results <- data.frame(m = m_samples, theta_sampled = m_sampled_values, error = m_sampled_errors
print(results)
```

```
##       m theta_sampled        error
## 1 1e+02     0.1238995 6.879827e-03
## 2 1e+03     0.1177041 6.844148e-04
## 3 1e+04     0.1165574 4.622859e-04
## 4 1e+05     0.1169331 8.649865e-05
```

```r
plot(log10(results$m), log10(results$error), type='b', xlab='log10(m)', ylab='log10(error)', ma
abline(lm(log10(results$error) ~ log10(results$m)), col='red')
```




```
slope <- coef(lm(log10(results$error) ~ log10(results$m)))[2]
cat("Slope of the line in log-log plot:", slope, "\n")
```

```
## Slope of the line in log-log plot: -0.5872113
```

■

2 Question 2

Let $X \sim \text{Exp}(1)$ and $Y = \mathrm{e}^{-X}$ so that $\theta = \mathbb{E}[Y] = \mathbb{E}[e^{-X}] = \int_0^\infty e^{-x}e^{-x}dx = 1/2$. Use $h(X) = X$ as a control variate (with $\mathbb{E}[h] = 1, \text{Var}(h) = 1$).

2.1 2a

Compute β^*, the optimal control coefficient.

□

$$\beta^* = \frac{\text{Cov}(Y, h)}{\text{Var}(h)} = \text{Cov}(e^{-X}, X) = \mathbb{E}[Xe^{-X}] - \mathbb{E}[X]\mathbb{E}[e^{-X}]$$

$$\mathbb{E}[Xe^{-X}] = \int_0^\infty xe^{-x}e^{-x}dx = \int_0^\infty xe^{-2x}dx = \frac{1}{4}$$

Thus, β^* is given by the following.

$$\beta^* = \mathbb{E}[Xe^{-X}] - \mathbb{E}[X]\mathbb{E}[e^{-X}] = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

■

2.2 2b

Compute $\text{Var}(Y)$, the minimized variance with control variate, and the relative efficiency.

□

For the $\text{Var}(Y)$, we can do the following.

$$\text{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 = \mathbb{E}[e^{-2X}] - \mathbb{E}[e^{-x}]^2 = \int_0^\infty e^{-3x}dx - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

The minimized variance with control variate is given by the following.

$$\text{Var}_{min} = \text{Var}(Y) - \frac{\text{Cov}(Y,h)^2}{\text{Var}(h)}$$

Thus, using the previously calculated values, we get the following.

$$\text{Var}_{min} = \frac{1}{12} - \left(-\frac{1}{4}\right)^2 = \frac{1}{12} - \frac{1}{16} = \frac{1}{48}$$

The relative efficiency is given by the ration of the variance to the minimized variance $\text{Var}(Y)/\text{Var}_{min}$. Thus,

$$\text{Relative Efficiency} = \frac{\text{Var}(Y)}{\text{Var}_{min}} = \frac{1/12}{1/48} = 4$$

■

3 Question 3

Let $\theta = \int_0^1 e^x dx = e - 1$ and $g(u) = e^u$.

3.1 3a

Derive $\mathbb{E}[g(U)]$ and $\mathrm{Var}(g(U))$ for $U \sim \mathrm{Unif}(0,1)$.

□

$$\mathbb{E}[g(U)] = \int_0^1 e^u du = e^1 - e^0 = e - 1$$

$$\mathbb{E}[g(U)^2] = \int_0^1 e^{2u} du = \frac{e^2 - 1}{2}$$

$$\mathrm{Var}(g(U)) = \mathbb{E}[g(U)^2] - (\mathbb{E}[g(U)])^2 = \frac{e^2 - 1}{2} - (e-1)^2 \approx 0.242$$

■

3.2 3b

Show that $g(U)$ and $g(1-U)$ are negatively correlated and conclude that the antithetic estimator has a smaller variance than the plain MC.

□

$$\mathbb{E}[g(U)g(1-U)] = \mathbb{E}[e^U e^{1-U}] = \mathbb{E}[e^1] = e$$

$$\mathrm{Cov}(g(U), g(1-U)) = \mathbb{E}[g(U)g(1-U)] - \mathbb{E}[g(U)]\mathbb{E}[g(1-U)]$$

Now, $\mathbb{E}[g(1-U)] = \int_0^1 e^{1-u} du = e - 1$, so we can write the covariance as follows.

$$\mathrm{Cov}(g(U), g(1-U)) = e - (e-1)^2 \approx -0.234$$

To compute the variance of the antithetic estimator, we can do the following.

$$\mathrm{Var}\left(\frac{g(U) + g(1-U)}{2}\right) = \frac{\mathrm{Var}(g(U)) + \mathrm{Var}(g(1-U)) + 2\mathrm{Cov}(g(U), g(1-U))}{4}$$

$$\mathrm{Var}\left(\frac{g(U) + g(1-U)}{2}\right) = \frac{e^2 - 1 - 2(e-1)^2 + 2e - 2(e-1)^2}{4} = \frac{e^2 - 1 + 2e - 4(e-1)^2}{4} \approx 0.0039$$

In comparison, we can compute the variance of the average of two variables with the same distribution.

$$\text{Var}\left(\frac{g(U)+g(U)}{2}\right) = \frac{\text{Var}(g(U)) + \text{Var}(g(U)) + 2\text{Cov}(g(U), g(U))}{4} = \frac{e^2 - 1 - 2(e-1)^2}{4} \approx 0.121$$

So we can conclude that the variance is lower when using the antithetic variables.

■

3.3 3c

With $m = 1000$ simulate both estimators $R = 100$ times and report the sample variances. Comment on the findings.

□

```r
function_3 <- function(x){
    return(exp(-x))
}

true_3 <- exp(0) - exp(-1)

m_3 <- 1e3
R_3 <- 1e2

# simple MC

simple_MC_3 <- function(m){
    running_sum <- 0

    for (i in 1:m){
        set.seed(i)
        x <- runif(1, 0, 1)
        running_sum <- running_sum + function_3(x)
    }

    return(running_sum / m)
}

simple_MC_values <- simple_MC_3(m_3)

# antithetic MC


antithetic_MC_3 <- function(m){
    running_sum <- 0

    for (i in 1:(m/2)){
        set.seed(i)
```

```
        u <- runif(1, 0, 1)
        running_sum <- running_sum + (function_3(1-u) + function_3(u))/2
    }

    return(running_sum / m)
}

antithetic_MC_values <- antithetic_MC_3(m_3)


simple_MC_variances <- numeric(R_3)
antithetic_MC_variances <- numeric(R_3)

for (r in 1:R_3) {
    simple_MC_variances[r] <- var(replicate(100, simple_MC_3(m_3)))
    antithetic_MC_variances[r] <- var(replicate(100, antithetic_MC_3(m_3)))
}
cat("Simple MC estimate:", simple_MC_values, "with variance:", mean(simple_MC_variances), "\n")
```

```
## Simple MC estimate: 0.6334347 with variance: 0
```

```
cat("Antithetic MC estimate:", antithetic_MC_values, "with variance:", mean(antithetic_MC_varia
```

```
## Antithetic MC estimate: 0.3161937 with variance: 0
```

As evident from the sample variances, we can see that the antithetic MC has a much lower variance by orders of magnitude. This points towards the usage of an anthithetic MC estimate to be much stronger.

■

4 Question 4

Let $U \sim \text{Unif}(0,1)$ and define $Y_1 = g(U)$ and $Y_2 = g(1-U)$ for a measurable $g : [0,1] \to \mathbb{R}$. Consider the antithetic estimator

$$\hat{\theta}_{\text{AV}} = \frac{1}{2m}\sum_{i=1}^{m}\left(g(U_i) + g(1-U_i)\right), \quad U_i \overset{\text{i.i.d.}}{\sim} \text{Unif}(0,1$$

4.1 4a

Show that $\hat{\theta}_{\text{AV}}$ is unbiased for $\theta = \mathbb{E}[g(U)]$

□

An estimator is unbiased iff $\mathbb{E}[\hat{\theta} - \theta] = 0$. We can show this by the following.

$$\mathbb{E}[\hat{\theta}_m] = \frac{1}{2m}\sum_{i=1}^{m}\mathbb{E}[g(U_i)] + \mathbb{E}[g(1-U_i)] = \frac{\mathbb{E}[g(U)] + \mathbb{E}[g(1-U)]}{2}$$

But, we know that since $U \sim \text{Unif}(0,1)$ then $\mathbb{E}[g(U)] = \mathbb{E}[g(1-U)] = \theta$.So we can simplify to the following.

$$\mathbb{E}[\hat{\theta}_m] = \frac{\mathbb{E}[g(U)] + \mathbb{E}[g(1-U)]}{2} = \frac{\theta+\theta}{2} = \theta$$

Thus the bias given by $\mathbb{E}[\hat{\theta}_m] - \mathbb{E}[\theta]$ is exactly 0.

■

4.2 4b

Let $\rho = \text{Corr}(g(U), g(1-U))$. Show that

$$\text{Var}(\hat{\theta}_{\text{AV}}) = \frac{1}{2m}\text{Var}(g(U))(1+\rho)$$

When does antithetic sampling reduce variance versus plain MC using 2m i.i.d. draws?

□

We have the following.

$$\text{Var}(\hat{\theta}_{\text{AV}}) = \text{Var}\left(\frac{1}{2m}\sum_{i=1}^{m}(g(U_i) + g(1-U_i))\right) = \frac{\text{Var}(g(U) + g(1-U))}{m}$$

Examining the inner variance, we get the following.

$$\text{Var}(g(U) + g(1-U)) = \frac{\text{Var}(g(U)) + \text{Var}(g(1-U)) + 2\text{Cov}(g(U), g(1-U))}{4}$$

But we know that $\text{Var}(g(U)) = \text{Var}(g(1-U))$. This allows us to rewrite as follows.

$$\text{Var}(g(U) + g(1-U)) = \frac{2\text{Var}(g(U)) + 2\rho\text{Var}(g(U))}{4} = \frac{\text{Var}(g(U))}{2}(1+\rho)$$

Thus, we can say that $\text{Var}(\hat{\theta}_{\text{AV}})$ is given by the following.

$$\text{Var}(\hat{\theta}_{\text{AV}}) = \frac{\text{Var}(g(U))}{2m}(1+\rho)$$

From the above, this will only reduce the variance if $\rho < 0$ which will lead to a decrease in the variance.

■

5 Question 5

5.1 5a

Suppose g is monotone on $[0,1]$. Determine the sign of $\rho = \mathrm{Corr}(g(U), g(1-U))$ for (i) monotone increasing g, (ii) monotone decreasing g. Give an intuitive explanation.

□

We know that the correlation is directly related to the covariance so to investigate the bahvior of ρ, it is sufficient to study the behavior of the covariance.

In other words, we need to show that the covariance is always negative regardless if g is monotonically increasing or decreasing. Without loss of generality, assume g is monotonically increasing. This can be said by the following.

$$x < y \implies f(x) \leq f(y) \quad \text{or} \quad f(y) - f(x) \geq 0$$

This in turns leads to the following statement.

$$x < y \iff 1 - x > 1 - y \implies f(1-y) - f(1-x) \leq 0$$

This means the product of the two preceding statements l(eads to the following.

$$\left(f(y) - f(x)\right)\left(f(1-y) - f(1-x)\right) \leq 0 \quad \forall x, y \in [0,1]$$

Take the double integral of this expression across the unit square.

$$I = \int_0^1 \int_0^1 (f(y) - f(x))(f(1-y) - f(1-x))dxdy$$

$$I = \int_0^1 \int_0^1 f(y)f(1-y) - f(y)f(1-x) - f(x)f(1-y) + f(x)f(1-x)dxdy$$

$$I = \int_0^1 \int_0^1 f(y)f(1-y)dxdy - \int_0^1 \int_0^1 f(y)f(1-x)dxdy - \int_0^1 \int_0^1 f(x)f(1-y)dxdy + \int_0^1 \int_0^1 f(x)f(1-x)dxdy$$

This can be simplified significantly by recognizing that the functions are symmetric and thus we can split this into the following sum, $I = I_1 + I_2 + I_3 + I_4$, and then analyze each part individually.

$$I_1 = \int_0^1 \int_0^1 f(y)f(1-y)dxdy = \int_0^1 f(y)f(1-y)dy$$

$$I_4 = \int_0^1 \int_0^1 f(x)f(1-x)dxdy = \int_0^1 f(x)f(1-x)dx$$

So we can say that by the proper change of variables, $I_1 = I_4$. We can write this form of the integral as follows.

$$I_a = \int_0^1 f(x)f(1-x)dx$$

Now taking I_2 and I_3, we see the following since the interiors are independent, we can seperate the integrals.

$$I_2 = \int_0^1 \int_0^1 f(y)f(1-x)dxdy = \left(\int_0^1 f(y)dy\right)\left(\int_0^1 f(1-x)dx\right)$$
$$I_3 = \int_0^1 \int_0^1 f(x)f(1-y)dxdy = \left(\int_0^1 f(x)dx\right)\left(\int_0^1 f(1-y)dy\right)$$

Which, by the proper change of variables again, we get the that $I_2 = I_3$.We can write this form of the integral as follows.

$$I_b = \left(\int_0^1 f(x)dx\right)\left(\int_0^1 f(1-x)dx\right)$$

Thus, we can write $I = 2I_a - 2I_b$, or less succinctly,

$$\frac{I}{2} = \int_0^1 f(x)f(1-x)dx - \left(\int_0^1 f(x)dx\right)\left(\int_0^1 f(1-x)dx\right)$$

Which the term $I/2$ is the covariance given by $\mathrm{Cov}(g(u), g(1-u))$. Thus we can say that regardless if the function g is monotonically increasing or monotonically decreasing, we get that the covariance is always negative.

■